Data Types and Features

Numerical, Categorical, Ordinal, Discrete, Continuous, Interval and Ratio

Class Notes · BCA-3004T Feature Engineering · Unit I

1. What is a Feature?

A feature is simply a characteristic or property describing an object. In Machine Learning, we select only those features that help prediction — this is the process of choosing important features and removing unnecessary ones from data. Good features improve machine learning performance.

However, features are not all of the same type. Some contain numbers, some contain labels, some contain rankings or grades, and some contain measurements. Before any feature engineering technique can be applied correctly, we must first learn to classify data correctly.

2. Why Data Type Matters: The Zip Code Trap

Consider two columns from a dataset: Age = 25 and Zip_Code = 201001. Both look like numbers, and both are stored as int64 in pandas. But can you take the average of Zip Codes? Does Zip_Code = 402002 mean 'twice as much location' as Zip_Code = 201001? No — that is meaningless. But Age = 50 genuinely is twice Age = 25.

> *Python does not know the meaning of your data — only you do.*

If the wrong data type is assumed, the entire feature engineering pipeline — scaling, encoding, aggregation, model choice — collapses. This is why data classification comes before any line of scikit-learn code is written.

> *"Discrete you count. Continuous you measure."*

3. The Classification Hierarchy

All data first splits into two broad families, and each family further branches into specific types:

- Numerical (Quantitative) — represents a measurable quantity
    - Discrete — countable, whole-number values
    - Continuous — can take any value within a range, including decimals
- Categorical (Qualitative) — represents groups or labels
    - Nominal — categories with no natural order
    - Ordinal — categories with a meaningful order

Numerical data itself is further classified by measurement level, moving from the least informative to the most informative scale:

- Nominal → name only, no order
- Ordinal → has rank, but gaps are not measurable
- Interval → equal gaps between values, but no true zero

- Ratio → equal gaps and a true, meaningful zero

4. Numerical (Quantitative) Data

Data that represents a measurable quantity — arithmetic operations on it are meaningful.

4.1 Discrete Data

Countable, whole-number values with gaps between them.

> *Can you have 2.5 children? No — you count children in whole numbers.*

Examples: Num_Children, Number_of_Products_Sold.

4.2 Continuous Data

Can take any value within a range, including decimals.

> *Can your height be 172.53 cm? Yes — height flows continuously.*

Examples: Height_cm, Weight_kg, Salary.

5. Categorical (Qualitative) Data

Data that represents groups or labels — arithmetic on it is meaningless.

5.1 Nominal Data

Categories with no natural order.

> *Is 'HR' greater than 'IT'? No — they are just different buckets.*

Examples: Gender, Department, Blood_Group.

5.2 Ordinal Data

Categories with a meaningful order, but the gap between categories is not measurable.

> *We know PhD > Bachelor. But is the 'distance' from High School to Bachelor the same as Bachelor to Master? We cannot say — that is the key difference from numerical data.*

Examples: Education_Level (High School < Bachelor < Master < PhD), Performance_Rating (Poor < Average < Good < Excellent).

6. Interval Data

Numerical data with equal gaps between values, but no true zero — zero does not mean "absence of the thing."

> *0°C does not mean 'no temperature' — it is just the freezing point of water. And 40°C is NOT 'twice as hot' as 20°C — that ratio claim is false.*

Another classic example is IQ Score:

> *An IQ of 150 does not mean someone is '1.5x as intelligent' as an IQ of 100. Zero IQ does not mean 'zero intelligence' — it is just an arbitrary scale point.*

Further examples reinforce the pattern:

- Calendar Years — the difference between Year 2000 and Year 2010 is a meaningful 10 years, but Year 0 does not mean 'time did not exist.'
- Floor Numbers in a Building — the difference between Floor 5 and Floor 1 is a meaningful 4 floors, but Floor 0 does not mean the building does not exist.

7. Ratio Data

Numerical data with equal gaps and a true, meaningful zero — so ratios such as 'twice as much' are meaningful.

> *Age = 0 genuinely means 'just born.' And Age 50 really is twice Age 25. This is what makes it Ratio, not Interval.*

Examples: Age, Height, Weight, Salary.

Everyday Examples of Ratio Data

Feature	What Does Zero Mean?
Money in wallet	No money
Mobile data balance	No internet data
Number of students	No students
Distance travelled	No distance
Salary	No salary
Number of books	No books
Time spent studying	No study time

Worked examples:

- Student A has ₹100 and Student B has ₹200 — B has twice as much money as A, and ₹0 means no money.
- Class A has 20 students and Class B has 40 students — Class B has twice the students, and 0 students means no students at all. This is ratio data.
- Daily mobile data usage of 1 GB, 2 GB, and 0 GB — 0 GB means no data used, and 2 GB is genuinely twice 1 GB.

8. Interval vs Ratio — Quick Comparison

Interval	Ratio
No true zero	True zero exists

Interval	Ratio
Example: Temperature (°C)	Example: Salary

9. Summary Comparison Table

Type	Order?	Equal Gaps?	True Zero?	Arithmetic Meaningful?	Example
Nominal	No	No	No	No	Gender, Department, Blood Group
Ordinal	Yes	No	No	No (ranking only)	Education Level, Performance Rating
Interval	Yes	Yes	No	Add / Subtract only	Temperature (°C), IQ Score
Ratio	Yes	Yes	Yes	Add / Subtract / Multiply / Divide	Age, Height, Salary, Weight

10. Common Trap to Remember

Zip_Code and Emp_ID look numeric (stored as int64) but are actually Nominal categorical data — no arithmetic mean of zip codes makes sense. This is the single most common real-world data type mistake, and a favourite examination or quiz question.

11. Key Takeaways

- A feature's data type must be judged by its meaning, not by how it is stored in pandas.
- Numerical data splits into Discrete (countable) and Continuous (measurable).
- Categorical data splits into Nominal (no order) and Ordinal (has order, unequal gaps).
- Interval data has equal gaps but no true zero; Ratio data has equal gaps and a true zero.
- Always ask: 'Does zero mean absence of the thing?' — the answer separates Interval from Ratio.
- Always ask: 'Is there a natural order?' and 'Are the gaps between values equal and measurable?' — these two questions place any feature correctly on the hierarchy.